Laser
A beam of light with a single wavelength, in which the waves are all perfectly in step. Lasers are used in electronics and surgery.

Milk is a type of **mixture**.

LED
Short for light-emitting diode. An LED is a device that produces light when an electric current passes through it. The colour of its light depends on the compounds used in it.

Liquid
A state in which the particles of matter (atoms or molecules) are only loosely attached to each other, and move freely. A liquid can flow and take any shape, but has a fixed volume.

Mass
The amount of matter in a substance.

Matter
The material that makes up everything around us.

Maglev
Short for magnetic levitation. This refers to some kinds of high-speed train that use magnets to propel themselves while hovering over a track.

Magnet
A solid object that produces a magnetic field, which attracts certain materials to it and can attract or repel other magnets.

Magnetic
Relating to a magnet.

Magnetic field
The force field around a magnet.

Melting point
The temperature at which a solid gets hot enough to turn into a liquid.

Metal
A type of element that is likely to react by giving away the outermost electrons in its atoms. Most elements are metals, and they tend to be hard, shiny solids. Mercury is the only metal that is liquid at room temperature.

Mineral
A naturally occurring solid compound – or mixture of compounds – made up of different elements. Every mineral has particular characteristics, such as crystal shape and hardness. Minerals are mixed together to make the rocks in Earth's crust.

Mixture
A collection of substances that fill the same space but are not connected by chemical bonds. Examples of mixtures are seawater, milk, and mud. The contents of a mixture can be separated by a physical process, such as filtration.

Molecule
A single particle of a compound. Its two or more atoms are bonded together.

Neutron
A neutral particle in the nucleus of an atom. A neutron is about the same size as a proton but it does not have an electric charge.

Noble gases
A group of elements that are unreactive and generally form no compounds with the other elements. This is because the outermost shells in their atoms are filled with electrons. This group sits on the extreme right of the periodic table.

Non-metal
A type of element that is likely to react by acquiring electrons in the outermost shell of its atoms. Non-metals are usually crystalline solids, such as sulfur, or gases, such as oxygen. Bromine is the only non-metal that is liquid at room temperature.

Nucleus
An atom's core, which contains its protons and neutrons. Nearly all the mass of an atom is packed into its nucleus.

Ore
A rock or mineral from which a useful element such as a metal can be purified and isolated.

Oxide
A compound in which oxygen is bound to one or more other elements.

Particle
A basic unit of which substances are made. Sub-atomic particles are units of which atoms are made including protons, neutrons, electrons, and many other smaller ones.

Particle accelerator
A machine in which atoms or sub-atomic particles are made to collide at high speeds. These collisions are then studied by scientists. Particle accelerators are used to produce artificial elements as well as study particles smaller than atoms. A cyclotron is a type of particle accelerator.

These petal-like shapes may form when desert sand mixes with barite, an **ore** of barium.